

Practice — Variables — What You Change vs What You Observe

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ**[1]**

In an experiment, the variable you **deliberately change** is called the —

- (a) measured variable (b) changed (independent) variable (c) controlled variable
(d) random variable

Q2 · MCQ**[1]**

A student changes the oil temperature and measures the time a puri takes to puff. The **measured** variable is —

- (a) oil temperature (b) dough thickness (c) time to puff (d) type of flour

Q3 · MCQ**[1]**

Which of these should be a **controlled** (kept same) variable when testing the effect of oil temperature?

- (a) the oil temperature (b) the thickness of each dough circle (c) whether the puri puffs
(d) the time to puff

Q4 · TRUE / FALSE**[1]**

It is fine to change two variables at the same time in one experiment.

- (a) True (b) False

Q5 · ASSERTION-REASON**[1]**

A: Controlled variables are kept the same throughout.

R: This makes sure only the changed variable can affect the result.

- (a) Both A and R true, and R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

The variable you observe or measure as the result is the _____ variable.

Q7 · FILL IN THE BLANK

[1]

"Did it puff: yes or no" is a _____ type of observation (one word).

Short answer (2 marks each)

Q8

[2]

Name the changed, measured and controlled variables for: "Does a longer ramp make a toy car roll further?"

Q9

[2]

Explain why we change only one variable at a time.

Q10

[2]

Give one example each of a qualitative and a quantitative measured variable for a plant-growth experiment.

Long / application (3 marks each)

Q11

[3]

Design a simple experiment to find out whether the **type of flour** (atta vs maida) affects how well a puri puffs. State the changed, measured and at least two controlled variables.

Q12

[3]

A friend says "hotter water always dissolves sugar faster" after one quick try where she also stirred faster in the hot water. Why is her experiment **not fair**, and how would you fix it?

[Check yourself](#) → Answer key